English Article Correction System Using Semantic Category Based Inductive Learning Rules

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Abstract. In this paper, we describe a system for automatic correction of English. Our system uses rules based on article context features, and generates new abstract rules by Semantic Category Based Inductive Learning that we proposed before. In the experiments, we achieve 93% precision with the best set of parameters. This method scored higher than our previous system, and is competitive with a related method for the same task.

1 Introduction

Using articles correctly is difficult for Japanese learners of English. Lee[1] reported that native speakers of languages which do not have any articles often have difficulty in choosing appropriate English articles, and tend to underuse them. Also Kawai et al.[2] reported there are many article errors in English written by Japanese.

In view of these circumstances, various methods for automatic detecting or correcting article errors have been proposed in the past. Kawai et al.[2] proposed a rule-based method which uses rules made by hand based on linguistic knowledge. To make such rules much effort and a lot of expenses are needed, and it is very difficult to achieve good coverage of articles usage. Izumi et al.[3] trained a maximum entropy classifier to recognize various errors including articles using contextual features. They reported the results for different error types (omission - precision 76%, recall 46%; replacement - precision 31%, recall 8%). Yi et al.[4] proposed a web count-based system for article error correction achieving precision 62% and recall 41%. De Felice et al[5] proposed a classifier-based approach to correct article and preposition errors (accuracy 92% for articles).

We also proposed an article error correction system[6] which corrects errors using rules extracted and generated automatically by Inductive Learning[7]. Apart from the performance, it has an advantage of being more understandable while investigating why a given article was erroneous because of using rules. However, its performance needs to be improved.

To improve the performance of our system, we propose an introduction of WordNet category information into rules for article error correction and Semantic

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Category Based Inductive Learning. Considering semantic category information, the learning process of our system can generate more reliable rules even though the system does not have any rules with the same head noun as user input. We aim to improve precision of article error correction.

The rest of the paper is organized as follows. In section 2, we describe details of the proposed system. In section 3, we evaluate the system and finally conclusions are given in section 4.

2 System

Fig. 1 shows the process flow of our system. When the system gets English sentences including article errors, it extracts feature slots. Using the resultant feature slots, the system extracts moderate amounts of rules from the rule database in order to generate abstract rules by Inductive Learning that is described in **2.3**. The system corrects article errors by the correction algorithm using the resultant rules. In the correction algorithm, the system calculates scores of rules to rank them by reliability.

We describe the details for each part of the process below.

2.1 Feature Slots and Rules

We use similar feature slots and rules as our previous system[6]. Consulting literature concerning usage of articles[8,9], we define feature slots that can be easily extracted automatically for the context within a sentence. A difference between the proposed system and our previously proposed system[6] is whether slots for nouns and verbs have WordNet category information or not. The proposed system extracts category information from WordNet 3.0 when it extracts feature

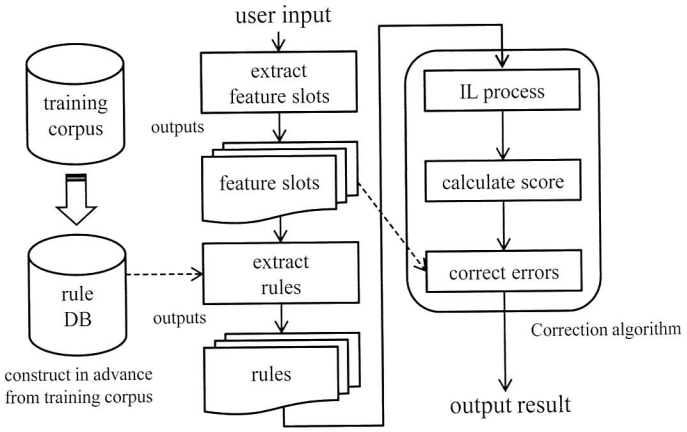
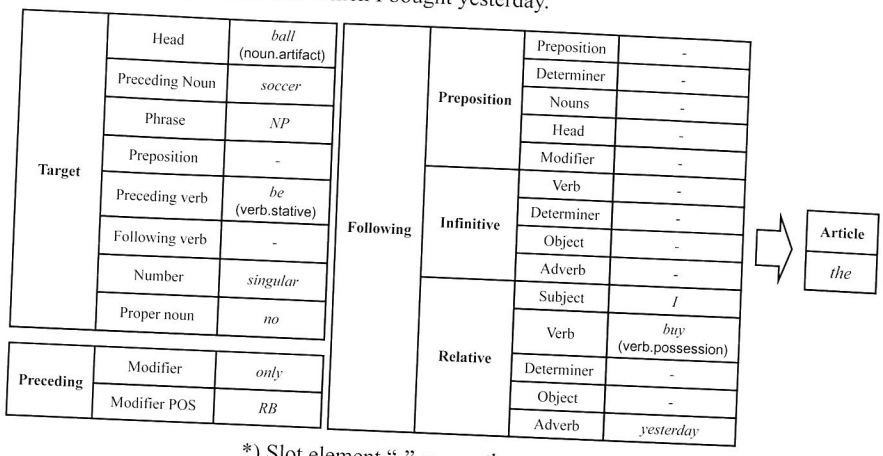


Fig. 1. Process flow

(i) This is the only *soccer ball* which I bought yesterday.



*) Slot element "-" means that no corresponding element is present.

Fig. 2. An example of a rule

slots. There are 26 semantic categories for nouns (e.g. 'noun.act', 'noun.food'), and 15 categories for verbs (e.g. 'verb.change', 'verb.creation'). A rule consists of a combination of an article and such feature slots. They are extracted from POS-tagged and chunked sentences automatically.

The feature slots and rule extracted from "soccer ball" in sentence (i) is as shown in Fig. 2. Feature slots consists of three categories, **Target**, **Preceding** and **Following**. **Target** category has context information about a target noun. Since "soccer ball" is a compound noun, "ball" is put into the *Head* slot and "soccer" is put into *Preceding Noun*. *Head* "ball" has category "noun.artifact" and *Preceding verb* has category "verb.stative". These categories are also put into *Head* and *Preceding verb* slots respectively. The **Preceding** category means modifiers preceding the target noun phrase, such as an adjective or adverb. The **Following** category contains information about modifiers following the target noun, such as prepositions, infinitive and relative clauses.

A requirement when applying rules to new text is that the rule feature slots and the context of the target noun must agree. If the slot element is "-", it counts as agreeing with any other element.

2.2 Rule Extraction

After extracting feature slots of user input, the system tries to extract rules that have feature slots similar to the user input from a rule database. These extracted rules are used in next learning process. The rule database is constructed from a training corpus in advance. The reason why we try to extract moderate amounts of rules for every user input is that it is impossible for the system to generate rules from the whole large corpus (say over 100 million words) because of the exponential growth of the recursive Inductive Learning.

When the system receives feature slots of user input, it searches rules from the rule database using queries based on feature slot elements of the input. The query patterns are as follows, in priority order:

1. **Target** *Preposition* and **Preceding** *Modifier* and **Target** *Head*
2. **Target** *Preceding Verb* and **Target** *Head* and **Following Preposition** *Preposition*
3. **Preceding** *Modifier* and **Target** *Head*
4. **Target** *Preceding Verb* and **Target** *Head*
5. **Target** *Preposition* and **Target** *Head*
6. **Target** *Head* and **Following Preposition** *Preposition*
7. **Target** *Head*
8. **Preceding** *Modifier*

If the system cannot extract rules by a query including **Target** *Head* or **Target** *Preceding Verb* element, it generates a new query that includes the semantic category information of the element instead of the element value. Using such queries with semantic category information, the system can get appropriate rules to be used in the learning process even though the rule database has no identical words as head nouns or verbs of user input.

2.3 Rule Abstraction

The system generates new abstract rules from extracted rules based on Inductive Learning (IL) that is proposed by us originally. IL is defined as discovering inherent regularities from actual examples by extracting their common and different parts recursively[7]. In our case, the actual examples are represented as the feature slots extracted from training data.

In the IL process, by extracting common and different elements recursively from comparison of feature slots of two rules, abstract rules are generated one after another. The abstract rules have the common parts of elements and another parts are abstracted.

However, by IL, abstract rules have only two kinds of elements, concrete element and full abstract element. Full abstract element is allowed to agree with any element. There is a possibility that many abstract rules which are too generalized and possess lower reliability are generated because of ease of full abstraction.

In this paper, we propose Semantic Category Based IL (SCB-IL). SCB-IL supplies additional granularity for rule abstraction, which abstract rules can have abstract elements with category information. We aim both to enhance accuracy of error correction and to carry on versatility of abstract rules using SCB-IL.

Fig. 3 shows an example of rule abstraction by SCB-IL. In Fig. 3, rule 1 is the same as Fig. 2 from sentence (i). Rule 2 and rule 4 are extracted from sentence (ii) and (iii) respectively.

Rule 3 is an abstract rule generated from rule 1 and rule 2. These two upper rules have many common slot elements: all of **Preceding**, a portion of **Target** and the article. The other elements differ. **Target** *Head* elements also differ, but

(i) This is the only *soccer ball* which I bought yesterday.
(ii) I have the only *guitar*.
(iii) Bobby is the only *child* in his family.

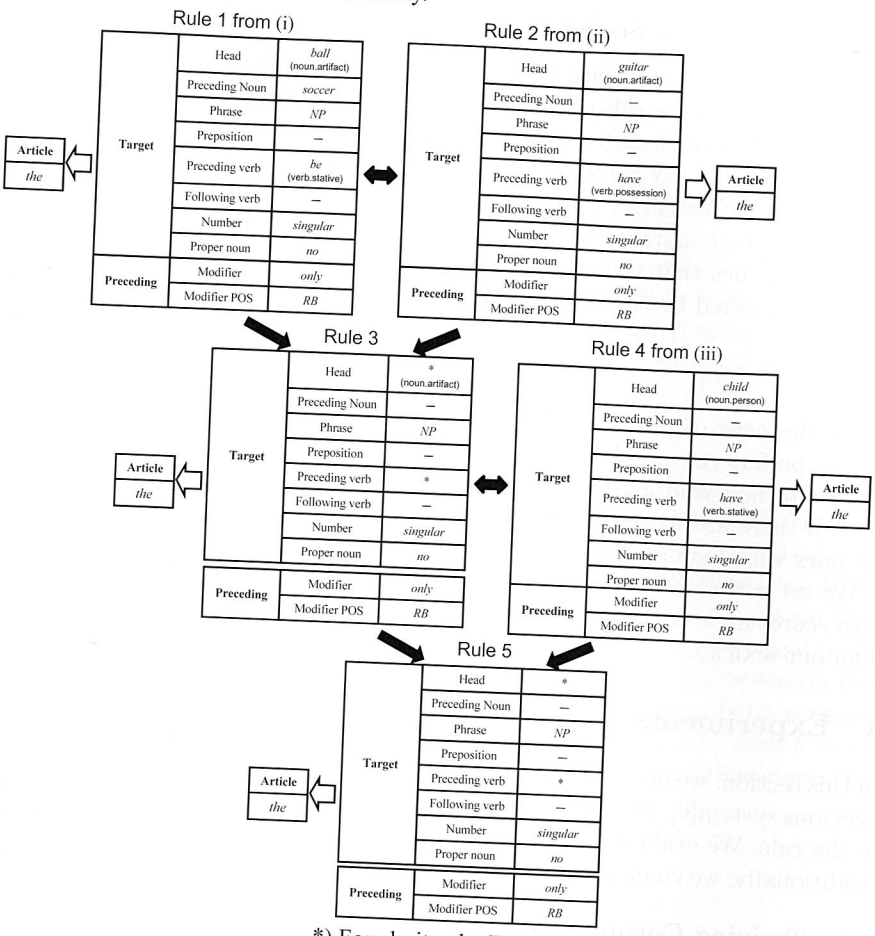


Fig. 3. An example of when the SCB-IL process generates new abstract rules

these categories are the same. SCB-IL thus generates rule 3 as a new abstract rule which has these common slot elements, an abstract element with category information and full abstract elements at the differing slots. If slots of source rules have different kinds of categories as **Target** *Head* slots of rule 3 and rule 4, the slot of the resultant rule is fully abstracted as rule 5. In Fig. 3, elements with "*" similarly to "-" mean the slot is allowed to agree with any element.

If the only requirement for SCB-IL was that two rules must have common slot elements, many rules abstracted too much would be generated. Thus, we add the requirements that the article elements of the rules must agree and that

Target *Head* or **Preceding** *Modifier* must also agree. If **Target** *Head* elements disagree but their categories agree, the system can continue SCB-IL. SCB-IL then generates rules recursively until no rules meeting these requirements remain.

2.4 Calculation of Scores for Rules

After preparing rules for error correction, the system calculates reliability scores for each rule using the rule database. This is because we assume rules in the rule database that is constructed from error free training corpus are very reliable.

Next we define "NA (Number of times Applied)" and "NAC (Number of times Applied Correctly)". NA means the number of times that a rule is applicable (context matches feature slots) in the rule database. NAC means the number of times that the rule is applicable and the article used is the same as the one suggested by the rule. The score of the rules is defined by the following formula:

$$score = \frac{NAC}{NA} \tag{1}$$

As in the case of our previous system[6], when correcting article errors, the system prefers rules with high specificity levels. Specificity level means the proportion of non-wild card elements (i.e., other than "-" and "*") to all feature slots. If there are some rules with the same specificity levels, the system prefers the ones with high scores.

We set a threshold parameter θ for the system. The system uses only rules with scores greater than or equal to θ. Therefore, the system may suggest more than one articles.

3 Experiments

In this section, we describe the evaluation experiment and a comparison with our previous system[6]. Our previous system[6] has no semantic category information in the rule. We evaluate effectivity of rules with semantic category and SCB-IL. Additionally, we compare the proposed system to a baseline and other methods.

3.1 Training Corpus and Test Data

We use the Reuters Corpus[10] (about 188 million words) as a source of the rule database. Constructing the rule database from the corpus in advance, it has 46,140,689 rules. We use Brill's Tagger[11] to extract feature slots from tagged sentences. The test data is sentences that include 2,269 article instances from the Reuters Corpus separate from the rule database. Since the test data does not include article errors, we evaluate whether the system suggests the same article as in the test data.

3.2 Experimental Procedures

We compare the proposed system and our previous system[6] results to the test data for $\theta = 0$ to 1 with 0.2 increments in between, and evaluate its precision (P) and recall (R) that are defined as follows:

$$P = \frac{\text{the number of suggesting articles correctly}}{\text{articles for which at least one rule matches the context}} \quad (2)$$

$$R = \frac{\text{the number of suggesting articles correctly}}{\text{all existing articles in the test data}} \quad (3)$$

And we also define F (F-measure) as follows:

$$F = \frac{2 \cdot P \cdot R}{P + R} \quad (4)$$

A requirement of suggesting articles correctly is that the rule with the highest score suggests the correct article.

Our proposed and previous systems need many rule extraction and abstraction processes if input has many noun phrases. Furthermore, these are not very lightweight processes. Therefore, in this experiment, we set 20 as the upper limit of the number of extracting rules from the rule database per noun phrase.

3.3 Results and Discussion

Fig. 4, 5 and 6 show the results of precision, recall and F-measure respectively. The threshold θ has a little effect on the change of the precision and recall of both systems, except for $\theta = 1.0$ in Fig. 5. The reason why only $\theta = 1.0$ degrades the recall is that there are very fewer rules with score 1.0 than others. In the course of the scores calculation, many rules have scores less than 1.0 even if the rule is reliable. Many reliable rules have scores close to 1.0 but not equal to 1.0.

Comparing both systems, the proposed system achieves better precision than the previous system in every θ as Fig. 4. The best precision of the proposed system is 0.93 with $\theta = 0.8$. That of the previous system is 0.87 with $\theta = 1.0$. On the other hand, the previous system has better recall performance than

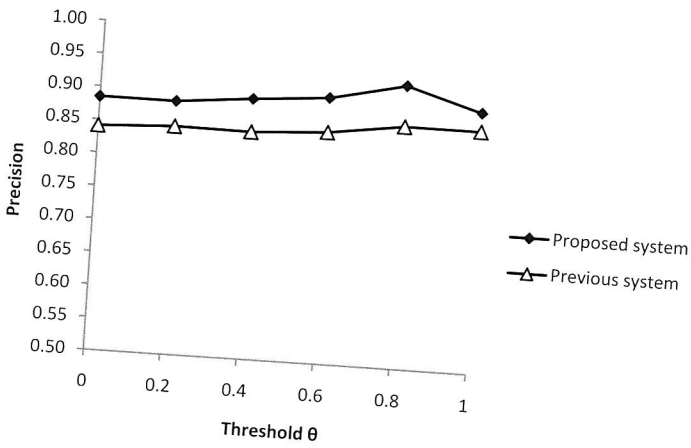


Fig. 4. Results of precision

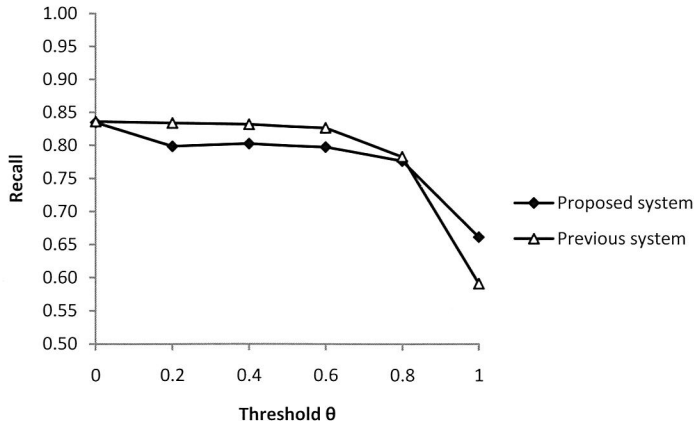


Fig. 5. Results of recall

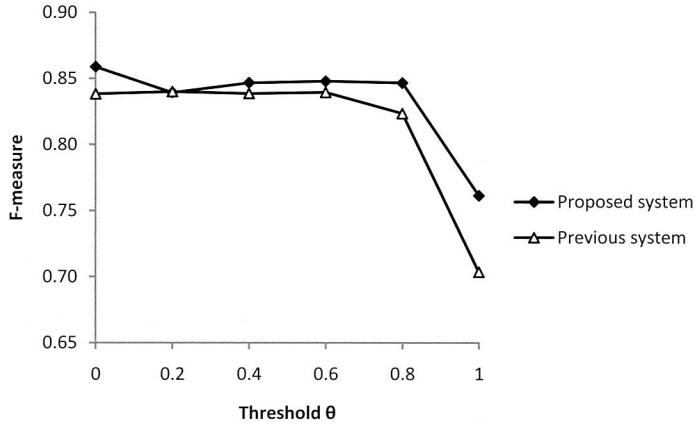


Fig. 6. Results of F-measure

the proposed system as Fig. 5. We consider the reason is the effect of SCB-IL which generates more specific rules than IL, as rules with semantic categories. Conversely, rules by IL are more general and easier to be applied to various contexts. However, the proposed system achieves better F-measure for about 2%. Therefore, the new method is more effective for error correction performance, especially accuracy.

Table 1 relates our results to other system[5] reported in the literature on comparable tasks for individual articles. In our system, it is not possible to find a correlation between the number of rules seen in rule database (**"a"**:8%, **"the"**:20%, **"null"**:72%) and precision and recall. We consider that the lowest performance for **"the"** is attributed to a lack of information of a rule. Choice of a definite article depends not only on context within a processed sentence but

Table 1. Individual articles results

System	Article	Precision	Recall
Baseline	**a**	-	0.00
	the	-	0.00
	null	0.72	1.00
De Felice et al.[5]	**a**	0.71	0.54
	the	0.85	0.92
	null	0.99	0.99
Our system ($\theta = 0.8$)	**a**	0.94	0.82
	the	0.88	0.69
	null	0.94	0.79

also on context of the preceding sentence or semantic information. A currently used rule has no information about the preceding sentence and sense except for WordNet category. We believe that the rule should have such information for performance improvement about definite articles.

In Table 1, The baseline refers to the most frequent article, **"null"**. Our system outperforms baseline in the comparison of precision. And in comparison of our system and De Felice et al.[5], our system gets over 20% better performance of both precision and recall for **"a"**. For other articles, our system can achieve similar precision to theirs. However, recall of our system is about 20% lower than theirs except for **"a"**. Our system suggests no article when it cannot generate rules that matches input context and we consider it as one of the factors of lower recall.

4 Conclusions and Future Work

In this paper, we proposed a system for correcting article errors based on automatic rules generation by Semantic Category Based Inductive Learning. Our proposed system achieved better precision performance than our previous system. We confirmed that the introduction of WordNet category information into rules for article error correction and SCB-IL have positive effect on our system.

However, our experiments are not sufficient in the following respects. First, the size of test data is too small. For example, De Felice et al.[5] used over 300,000 article contexts for test data. Second, our experiments have no error correction for sentences that are written by second learners of English. We try to evaluate the system with larger test data and with sentences including article errors by the second learners of English in the near future.

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